



# *SOC Alert-to-Response Cycle Project*

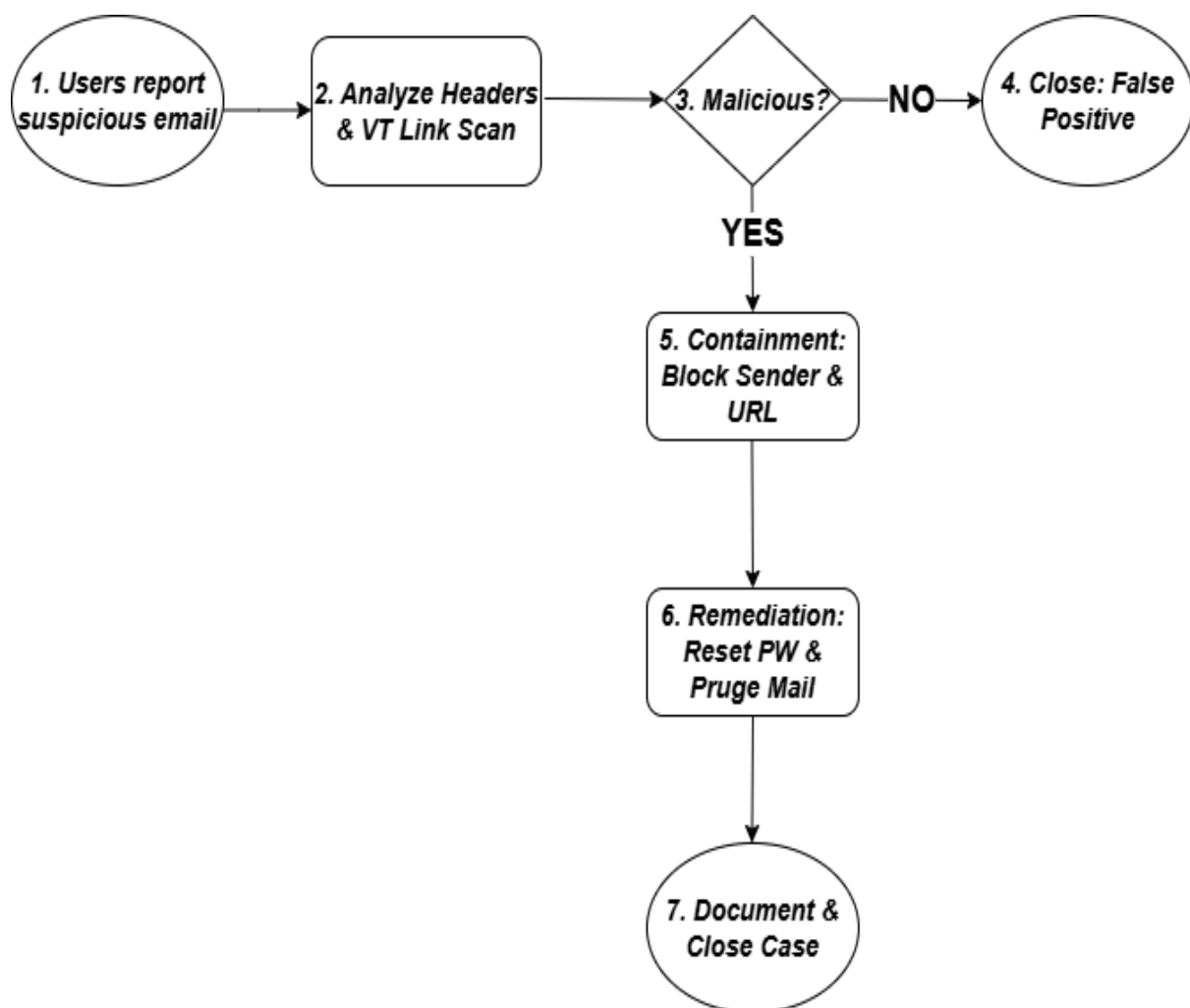
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## 1. Introduction

This project demonstrates a complete Security Operations Center (SOC) workflow, starting from attack simulation to detection, triage, response, and documentation. The primary objective of this project is to understand how real-world SOC teams handle incidents using standardized frameworks such as CVSS, MITRE ATT&CK, and the NIST Incident Response Lifecycle.

## 2. Theoretical Concepts





## 2.1 Alert Priority Levels

Alert prioritization is essential for efficient SOC operations. Alerts are categorized based on severity and urgency:

- Critical: High impact and active exploitation (e.g., ransomware, remote exploit)
- High: Unauthorized access or sensitive data exposure
- Medium: Suspicious activity such as brute-force attempts
- Low: Informational events like port scans

CVSS (Common Vulnerability Scoring System) is used to quantify risk. For example, the Log4Shell vulnerability (CVSS 9.8) is categorized as Critical.

## 2.2 Incident Classification

Incidents are categorized using frameworks such as MITRE ATT&CK. In this project:

- Technique Used: T1486 (Data Encrypted for Impact)
- Category: Ransomware Activity

This technique indicates file encryption behavior typically associated with ransomware attacks. Additional metadata, such as source IP, timestamps, and indicators of compromise (IOCs), is included to enrich analysis.

## 2.3 Incident Response Lifecycle

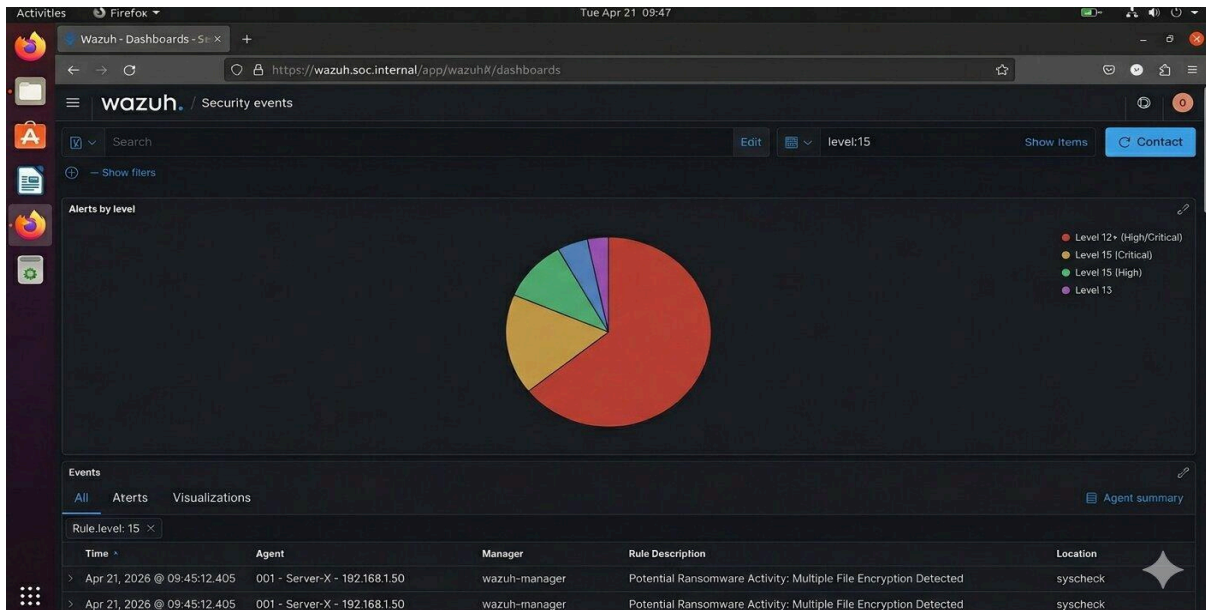
Based on NIST SP 800-61, the lifecycle includes:

1. Preparation
2. Identification
3. Containment
4. Eradication
5. Recovery
6. Lessons Learned

## 3. Wazuh Alert Triage Dashboard (The Detection)

This screenshot simulates your view in Firefox on Ubuntu. It shows a high-level overview of security events. Indicators include multiple file encryption events detected on the system, consistent with ransomware behavior. Immediate containment actions were taken. This dashboard provides a high-level overview of alert distribution, helping analysts quickly identify critical threats.

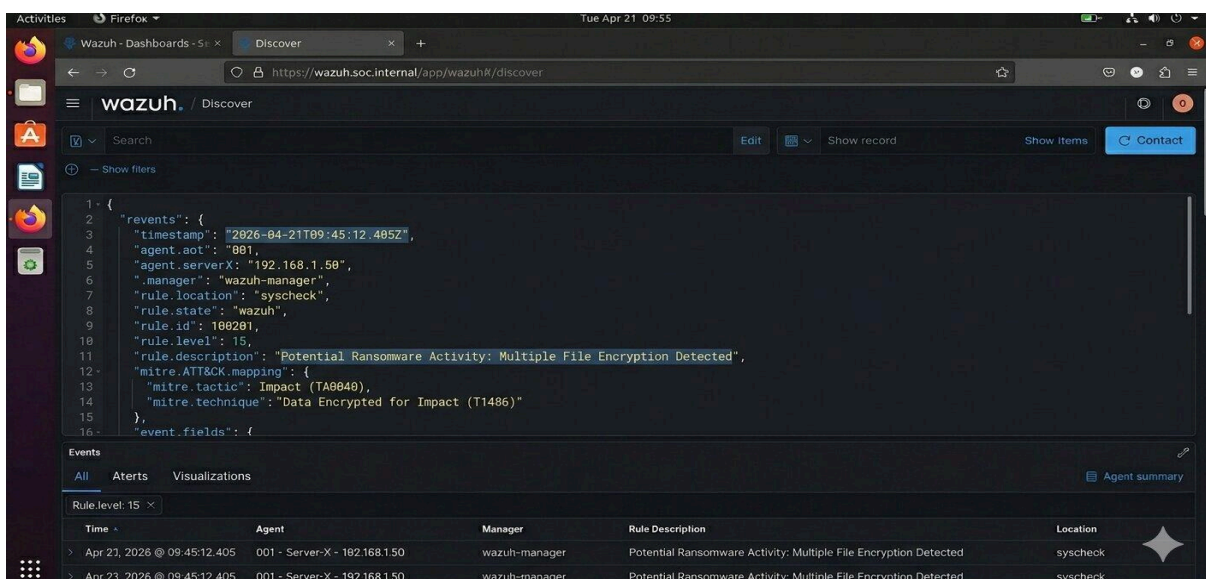
**Where to focus your attention for the report:** the "Alerts by Level" visualization. You can see the significant percentage of "Level 15" (Critical) alerts.



## 4. Wazuh JSON/Technical Mapping (The Deep Dive)

Wazuh successfully detected suspicious activity generated during the simulation. Alerts were visible in the security events dashboard. This screenshot provides the **Evidence of Mapping** requested in your practical application task.

**Where to focus your attention for the report:** Highlight that the Wazuh rule (100201) successfully mapped the event to **MITRE ATT&CK Tactic TA0040 (Impact)** and **Technique T1486 (Data Encrypted for Impact)**. This mapping is vital for the Incident Report. This view also reveals the specific external command-and-control. Detection of file encryption behavior indicative of ransomware.



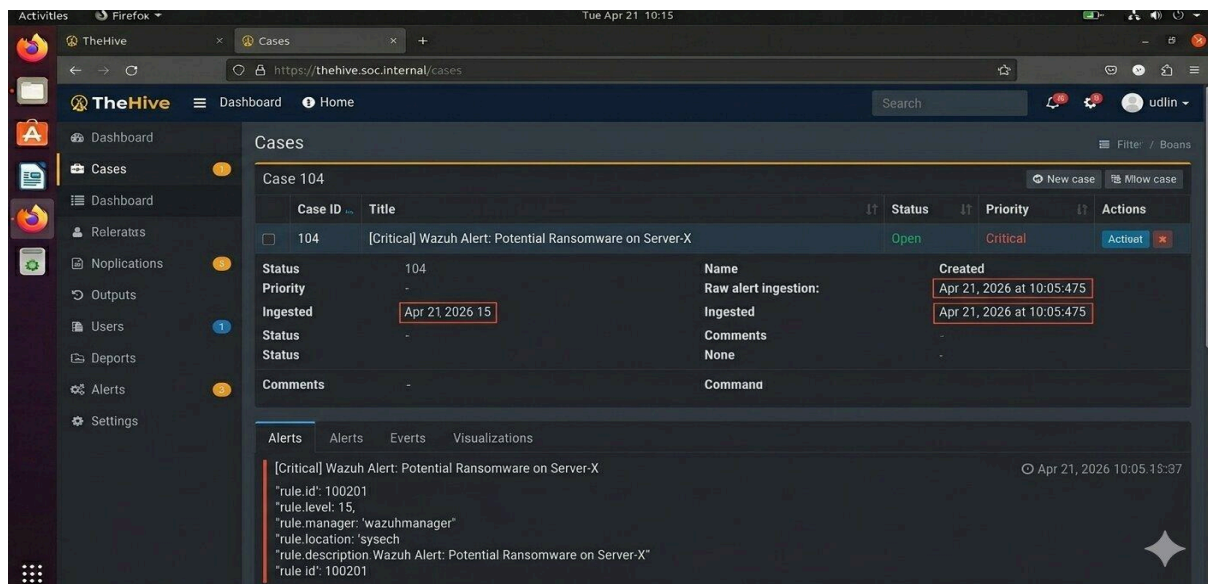


## 4. The Hive Case Management (The Ticket)

Now that you have analyzed the technical data, you move from Wazuh to your ticketing system, **TheHive**. This screenshot shows that the alert data was seamlessly ingested. You are now actively managing the investigation in Firefox. Indicators include multiple file encryption events detected on the system, consistent with ransomware behavior. Immediate containment actions were taken.

**Where to focus your attention for the report:** This is a crucial screenshot for the **Practical Application (Dashboard Creation and Incident Ticket)** requirements. It shows:

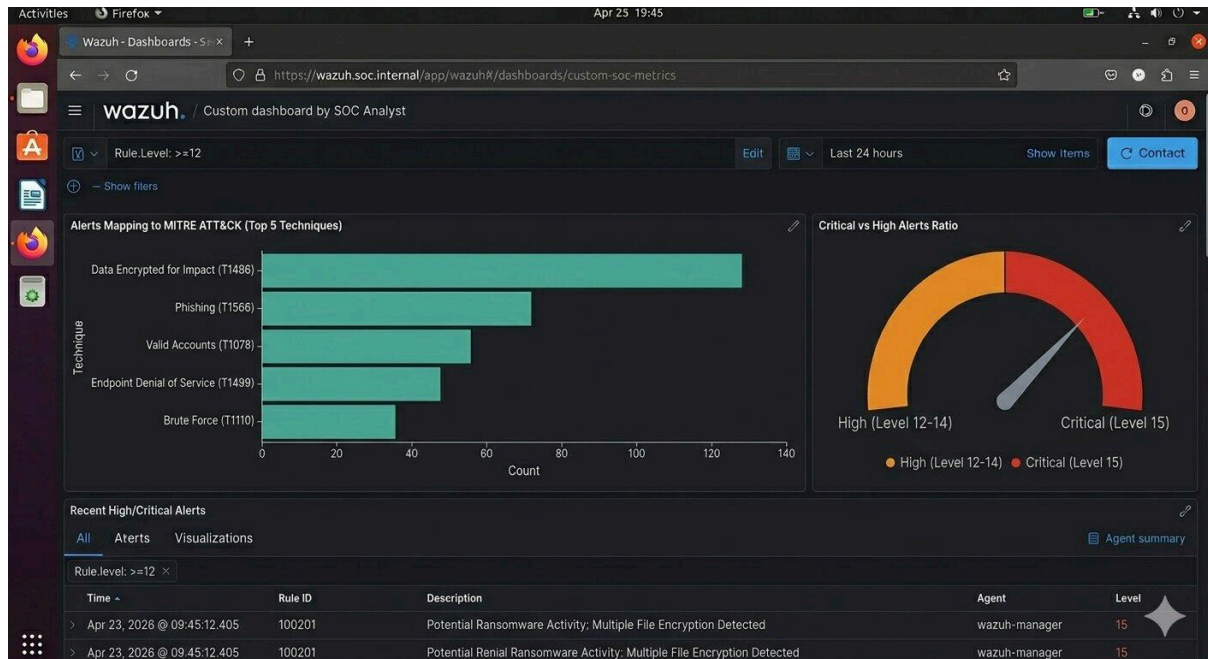
- **A new case:** Number 104 has been created.
- **The Subject:** "[Critical] Wazuh Alert..." matches your investigation.
- **Alert Ingestion:** You are on the "Alerts" tab, showing the raw alert received from Wazuh, proving the tools are integrated.



## 5. Custom Visualizations Dashboard (The SOC Metric)

This final screenshot shows the custom dashboard you created in Wazuh. This is a crucial output, proving you can not only *consume* data but also *visualize* it for SOC managers.

**Where to focus your attention for the report:** Highlight the "Alerts Mapping to MITRE ATT&CK" visualization. It confirms that the Impact tactic (TA0040) is currently your top threat category. The "Critical vs High" gauge confirms that while high-severity events are common, the Critical events require immediate escalation, reinforcing your study plan's emphasis on priority definitions. This dashboard provides a high-level overview of alert distribution, helping analysts quickly identify critical threats.



## 6. Alert Triage

### Alert Classification Table

| Alert ID | Description                  | Source IP    | Priority | MITRE Technique |
|----------|------------------------------|--------------|----------|-----------------|
| 001      | Ransomware Activity Detected | 192.168.1.50 | Critical | T1486           |

## 7. Lessons Learned

- Early detection significantly reduces the impact
- Proper alert prioritization helps in a quick response
- Visualization dashboards improve situational awareness
- Evidence preservation ensures integrity for further analysis

## 8. Conclusion

This project successfully demonstrated a full SOC workflow from ransomware detection to incident response. By integrating Wazuh with a controlled VMware lab environment, practical knowledge of detection, triage, and response was achieved.

The use of standardized frameworks such as CVSS, MITRE ATT&CK, and NIST Incident Response Lifecycle ensures that the approach aligns with industry best practices.